The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, SPRING SEMESTER 2017-2018

INTRODUCTION TO SOFTWARE ENGINEERING

Time allowed: ONE HOUR

ANSWER

Candidates must NOT start writing their answers until told to do so

Answer ALL of the Questions

Only silent, self-contained calculators with a single-line display are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

SECTION A [25 Marks]

Software Development Scenario

Your startup company have received a project from LRT Putrajaya about handling issues on ticket processing system. The scenario for this project is as follow: A customer service technician receives a telephone call, email, or other communication from a customer about a problem. Some applications provide built-in messaging system and automatic error reporting from exception handling blocks. The technician verifies that the problem is real, and not just perceived. The technician will also ensure that enough information about the problem is obtained from the customer. This information generally includes the environment of the customer, when and how the issue occurs, and all other relevant circumstances. The technician creates the issue in the system, entering all relevant data, as provided by the customer. As work is done on that issue, the system is updated with new data by the technician. Any attempt at fixing the problem should be noted in the issue system. Ticket status most likely will be changed from open to pending. After the issue has been fully addressed, it is marked as resolved in the issue tracking system. If the problem is not fully resolved, the ticket will be reopened once the technician receives new information from the customer.

1. Answer the following questions using the scenario given above.

[Bloom's Taxonomy: Knowledge]

a. What is the difference between class diagram and state machine diagram? [2 Marks]

ANSWER for a: (one mark for class diagram and one mark for state machine diagram)

A class diagram shows classes in their relation and their properties and methods.
A state diagram visualizes a class's states and how they can change over time.

[Bloom's Taxonomy: Knowledge, Comprehension]

b. Explain what is Use Case diagram? How use cases can help analysis team and development team in the project (2 reasons for each). [5 marks]

ANSWER for b: (one mark for use case definition, 2 mark for analysis team and 2 marks for development team)

Use case diagrams are usually referred to as behavior diagrams used to describe a set of actions (use cases) that some system or systems (subject) should or can perform in collaboration with one or more external users of the system (actors).

Use cases help the analysis team:

- To improve communication among the team members
- To encourage common agreement about system requirements

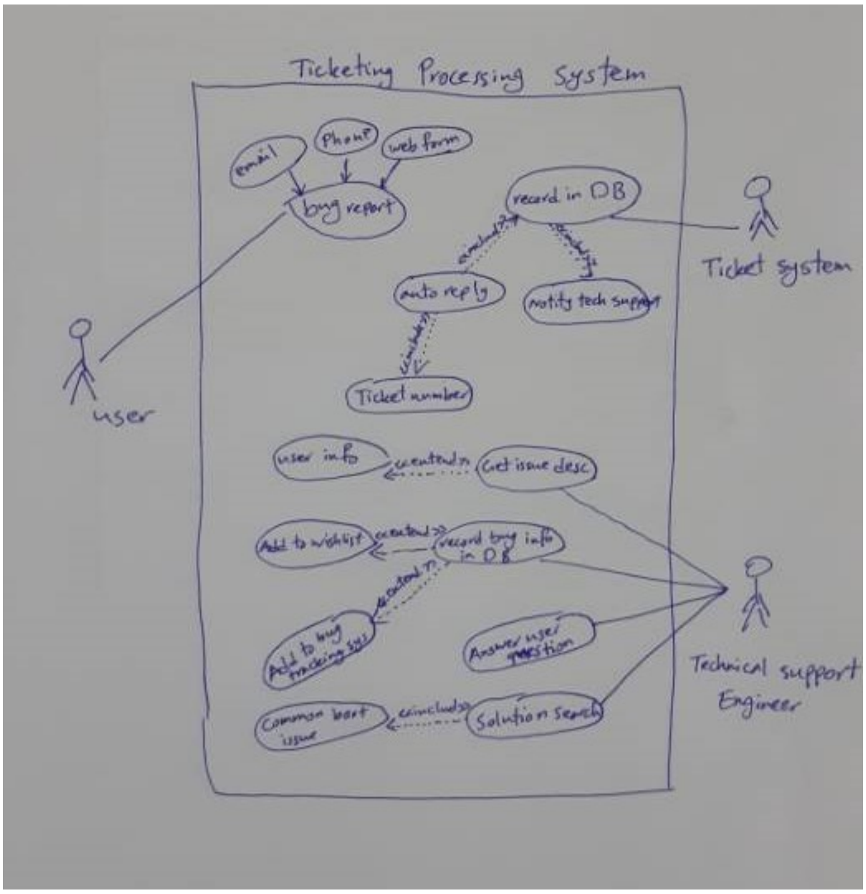
In addition, use cases help the development team:
- To understand the business processes
- To recognize patterns and contexts in functional requirements.

[Bloom's Taxonomy: Comprehension, Application]
c) Draw a use case diagram for the given above scenario. [7 Marks]

ANSWER for c:

7 marks for the correct use case diagram.

1 mark for actor, 2 marks for use cases, 1 mark for arrow, 3 marks for included and extended



[Bloom's Taxonomy: Knowledge]
d) List out 3 benefits of using sequence diagram in your design stage of your software development. [3 Marks]

ANSWER for d: (one mark for each benefit, max 3 marks)

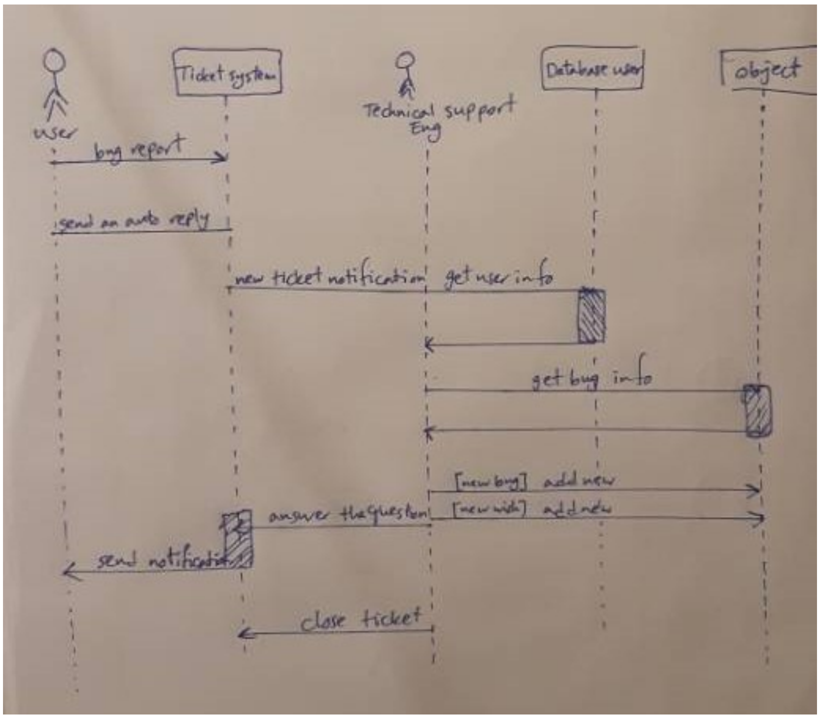
Sequence diagram can help the team to discover architectural, interface and logic problems early. In addition, sequence diagrams allow you to discuss the design in concrete terms. Sequence diagram also can be used to document the dynamic view of the system design at various level of abstraction.

[Bloom's Taxonomy: Comprehension, Application]

e) Draw a sequence diagram for the given above scenario. [8 Marks]

Max 8 marks for the correct sequence diagram.

2 marks for the objects, 2 marks for correct activation, 2 marks for correct sequence and 2 marks for the right message.



SECTION B [25 Marks]

2.

[Bloom's Taxonomy: Knowledge]

a) State the advantages of using version control when working in a software development team that is distributed over multiple geographical locations. [4 Marks]

ANSWER for 2a: (each advantage 1 mark, max 4 marks)

- Version control provides consistent methods for an entire team to use with each member of the team operating under the same 'ground rules'.
- Changes are orderly – with confusion being avoided
- The ability to track changes promotes accountability and makes it easier to find the right person to solve problems in the materials maintained.
- A list of exact changes made can be generated quickly and easily, making it easier to advise users of the information on how it has changed from version to version.
- It is easy to 'roll back' to an earlier version of the code, if errors are discovered.

[Bloom's Taxonomy: Comprehension, Application]

b) list 5 functional requirements for the cash-dispensing function of a bank ATM machine. [5 marks]

ANSWER for 2b: (max 5 marks for 5 functional requirements)

The system must provide a facility, which allows a specified amount to cash to be issued to customers. The customer requests the amount but the system may reduce this amount if the customer's daily limit or overdraft limit is reached.
The sequence of actions to dispense cash should be:
1. The customer inputs the amount of cash required
2. The system checks this against daily card limits and the customer's overdraft limit.
3. If the amount breaches either of these limits, then a message is issued which tells the customer of the maximum amount allowed and the transaction is cancelled.
4. If the amount is within limits, the requested cash should be dispensed
5. The customer's account balance and daily card limit should be reduced by the amount of cash dispensed.

[Bloom's Taxonomy: Knowledge, Comprehension]
c)

i) Compare and contrast the Waterfall design methodology with eXtreme Programming (XP). [4 Marks]
ii) What are the benefits and drawbacks of both approaches to software design? [2 Marks]

ANSWER for 2c:

1 mark will be awarded for each relevant point made, depending upon how important it is and how clearly it is stated.

i)

- Waterfall focuses upon organizing and completing one phase and then fully transitioning to the next. XP plans for change
- Waterfall projects start with a single requirements phase, that is then translated into program design. XP employs an iterative approach, with the iterations made visible to the customer.
- With waterfall there is a major software release at the end of the project, with XP releases are incremental.
- With waterfall the customer doesn't use the software until the end of the process, with XP the customer is involved throughout.
- With waterfall validation is done during acceptance testing, with XP it is done as the project progresses.
- With waterfall changes in requirements are difficult and expensive, with XP they are much easier and cheaper.

ii)

- Waterfall aims to get the design right, and then deliver what the customer needs. XP focuses upon getting some key functionality to the customer quickly – even if this has problems, and then fixing those problems.
- Coding for waterfall is based upon design and then test. With XP testing is much more at the forefront – with test units released at short intervals, sometimes on a daily basis.

[Bloom's Taxonomy: Knowledge, Comprehension]

d)

i) Describe the Scrum development framework. [3 Marks]
ii) Why isn't Scrum used for all software development? [3 Marks]

ANSWER for 2d: (3 marks for the first part and 3 marks for the second part)

i)
Scrum is a team--based agile framework (named after a rugby scrum, where the ball is driven forward by the team as a whole). The main concepts of Scrum are: product backlog (a prioritised list of requirements – expressed as user stories); the sprint (a mini 1--4-week project); a cross--functional team of 5--9 people; the product owner (who is responsible for the backlog); and the scrum master (who facilitates the process and removes impediments). There is no conventional project manager. A burn--down chart consisting of the backlogs not yet implemented in the current sprint is always publicly displayed. There is a daily scrum – a 15-minute meeting where everyone states "what they did yesterday", "what they will do today" and "what impediments there are in their work".

ii)
It does not always sit well with large corporations and large projects. The lack of upfront design and documentation can make contract negotiations hard. It assumes empowered knowledge workers and extremely good communication within the organisation. If is not the case, then Scrum projects tend to fail. They only work with co -located teams – given the emphasis on face to face contact it does not work well with distributed teams. Heavily documented legacy projects can be very hard to fit into a Scrum framework.

[Bloom's Taxonomy: Knowledge]

e. Explain the difference between "requirements definition" and "requirements specification"? [4 Marks]

ANSWER for 2e: (2 marks for requirement definition and 2 marks for requirement specification)

Requirements definition: A statement in natural language plus diagrams of the services the system provides and its operational constraints. Written for customers. Requirements specification: A structured document setting out detailed descriptions of the system services. Written as a contract between client and contractor. Maximum of 3 marks for each definition.

THE END